LYNN MARGULIS

1938–2011

American biologist Margulis attended Chicago University aged only 15 and gained her Ph.D. at the University of California, Berkeley, in 1965. The next year, at Boston University, she proposed that cells within nuclei had evolved as a result of the symbiotic merger of bacteria. This idea, although not generally accepted until the 1980s, transformed the understanding of cell evolution.

See also: The Gaia hypothesis

PAUL F. HOFFMAN

1941–

Canadian scientist Paul Hoffman's discovery of "cap carbonates"—evidence for ancient glaciation in Precambrian sedimentary rocks in Namibia—revived the "Snowball Earth" hypothesis in climate change studies in 2000. The term was first used by American geologist Joseph Kirschvink in 1992, although there had been speculation since the late 19th century that Earth's surface was almost entirely frozen more than 650 million years ago.

See also: Ancient ice ages

SIMON A. LEVIN

1941–

Levin, an American ecologist, specializes in the use of sophisticated mathematical modeling, alongside field and lab observation, to understand the workings of ecosystems. He also researches the relationships between ecology and economics. Levin earned a Ph.D. in mathematics from the University of Maryland in 1964 and taught at Cornell University from 1965